

Tarun Gupta

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EDUCATION

Toyota Technological Institute at Chicago (TTIC), USA
Ph.D., Computer Science, 2026–2031 (expected)

Indian Institute of Technology (IIT) Indore, India
B.Tech., Computer Science and Engineering, 2018–2022
CGPA: 9.81/10

PUBLICATIONS

[C1] Beyond World Models: Rethinking Understanding in AI Models. AAAI 2026
Tarun Gupta, Danish Pruthi.

[C2] All That Glitters is Not Novel: Plagiarism in AI Generated Research. ACL 2025 (Outstanding Paper Award)
Tarun Gupta, Danish Pruthi.

[C3] Improving Spoken Language Identification with Map-Mix. ICASSP 2023.
Shangeth Rajaa, Kriti Anandan, Swaraj Dalmia, **Tarun Gupta**, and Chng Eng Siong.

[C4] Estimation of Speaker Age and Height From Speech Signal Using Bi-Encoder Transformer Mixture Model. Interspeech 2022.
Tarun Gupta, Duc-Tuan Truong, Tran The Anh, Chng Eng Siong.

[J1] Self-supervised learning for macromolecular structure classification based on cryo-electron tomograms. Frontiers in Physiology.
Tarun Gupta, Xuehai He, Mostofa Rafid Uddin, Xiangrui Zeng, Andrew Zhou, Jing Zhang, Zachary Freyberg, and Min Xu

EXPERIENCE

Research Associate, Indian Institute of Science Jul 2024 – Present

- Investigated plagiarism in AI research agents and discovered that 24% of AI-generated research documents plagiarize existing work without attribution, evading current detection systems [C1].
- Examined the limitations of “world models” as a framework for characterizing human-level understanding through case studies inspired from philosophy of science and epistemology [C2].
- [Exploratory; unsuccessful] Attempted to model synaptogenesis and pruning via filopodia within mathematical brain model NEMO [Dabagia et al., 2024], inspired by recent findings on its unexpected prevalence in adult brains [MIT News, 2022].
- [Exploratory; unsuccessful] Investigated whether co-reference chain heads’ concentration in document beginnings causes primacy bias in LLM attention patterns, contributing to the lost-in-the-middle problem [Liu et al., 2023].

Senior Member Technical, D. E. Shaw (Hedge Fund) Jul 2022 – Jun 2024

- Developed distributed job scheduling software for managing large-scale ML training workloads, addressing challenges in resource allocation, scalability, and system efficiency. Project featured in Business Insider [Link].
- Promoted to Senior Member Technical (L2) in just 1.5 years, an achievement earned by around 10% of eligible staff.

Research Intern, NTU Singapore & Skit AI Dec 2021 – Jun 2022

- Built a mixture-of-experts (MoE) transformer model for speaker profiling, leading to a relative improvement of 18.5% in male age estimation and 8.6% in female age estimation on the TIMIT dataset [C4].
- Developed Map-Mix, a data augmentation method for spoken language identification that leverages training dynamics to improve sampling for latent mixup,

improving weighted F1 by 2% over random mixup [C3].

Research Intern, Computational Biology Dept., CMU Jan 2021 – May 2021

- Built a contrastive self-supervised learning pipeline for macromolecular structure classification using cryo-electron tomography data; extended SimCLR, MoCo, and SwAV to the cryo-ET domain, achieving significant gains over non-self-supervised baselines [J1].

Research Intern, Mathematics Dept., IIT Indore Jan 2020 – Dec 2020

- Performed research in non-convex optimization, focusing on SVM variants and clustering methods.

HONORS & AWARDS

Outstanding Paper Award, ACL 2025 (top 1% of accepted papers).

Best Undergraduate Researcher Award, IIT Indore, 2021.

Department Rank 1 at IIT Indore (based on CGPA).

Silver medal, Inter IIT Tech Meet 9.0 (Bridgei2i NLP Competition), 2021.

Bronze medal, Inter IIT Tech Meet 8.0 (BITGRIT Data Science), 2020.

Recipient of Indian Academy of Sciences Summer Research Fellowship, 2020.

TEACHING EXPERIENCE

Teaching Assistant, Indian Institute of Science Jan 2025 – Apr 2025

- Supported the Introduction to NLP course by conducting tutorial sessions, grading assignments, and providing one-on-one assistance to students.

Head, ML Student Club, IIT Indore Aug 2020 – Jun 2021

- Organized tutorials and competitions, and mentored student projects in machine learning.